

WILLIAMSBURG, SC, USA

WMO: 720604

Lat: 33.717N

Lon: 79.850W

Elev: 66

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00196

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	27.9	9.4	8.9	33.1	14.0	11.2	37.0	18.6	52.7	16.8	53.4	1.2	290	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	95.5	77.4	93.0	76.7	90.8	76.2	83.1	90.9	81.3	86.6	79.9	85.2	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.4	163.5	86.3	79.4	153.2	83.5	78.8	149.6	82.8	46.9	89.0	44.8	85.7	43.3	85.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.8	12.5	10.7	18.0	98.1	4.6	3.5	14.7	100.6	12.0	102.6	9.5	104.6	6.2	107.1		
			16.8	83.4	4.3	2.5	13.7	85.3	11.2	86.7	8.8	88.2	5.7	90.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	46.3	50.8	55.2	64.1	72.6	79.5	81.1	79.6	75.2	65.4	53.9	50.0
	DBStd	14.34	10.21	9.08	9.25	7.34	5.75	4.09	3.47	3.13	4.58	7.55	8.04	9.45
	HDD50	508	194	93	53	2	0	0	0	0	0	2	44	120
	HDD65	2338	583	405	324	104	13	0	0	0	2	90	344	473
	CDD50	5820	79	115	214	426	700	886	965	917	757	479	161	121
	CDD65	2174	3	7	21	77	248	436	500	452	309	101	11	9
	CDH74	20366	23	58	225	820	2256	4442	5070	4057	2496	814	74	31
	CDH80	8454	0	2	35	201	860	2073	2375	1735	974	194	4	1
Wind	WSAvg	3.5	3.7	4.4	4.7	4.7	3.9	3.2	3.2	2.4	2.6	2.7	2.8	3.2
Precipitation	PrecAvg	50.5	4.4	3.7	4.4	3.1	3.7	5.3	5.2	6.2	4.3	3.4	2.6	3.5
	PrecMax	72.3	8.3	9.6	9.2	9.3	11.4	14.1	12.3	12.8	13.5	12.6	7.1	10.0
	PrecMin	36.3	1.3	1.0	0.9	0.3	0.8	1.5	0.4	1.8	0.0	0.0	0.6	0.7
	PrecStd	7.7	1.9	1.9	2.1	2.0	2.0	2.9	2.7	2.7	2.8	3.2	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	79.5	83.8	88.1	95.1	99.2	99.4	95.3	92.9	88.5	80.8	77.4
		MCWB	67.4	65.2	67.3	70.9	71.7	82.0	81.9	78.4	74.5	73.8	69.4	68.0
	2%	DB	71.7	75.1	80.7	85.7	90.7	95.4	96.7	92.6	90.5	85.3	76.5	72.9
		MCWB	64.5	64.0	65.3	68.3	71.0	76.9	79.0	77.4	74.9	72.0	66.7	67.0
	5%	DB	67.8	72.0	76.6	81.8	88.3	91.4	93.1	90.6	88.4	81.6	72.4	70.1
		MCWB	60.7	61.4	62.5	66.5	71.2	76.1	78.0	77.6	74.2	70.3	64.7	65.6
	10%	DB	63.4	67.8	72.1	79.2	85.7	90.3	90.5	89.6	86.2	79.0	69.7	66.2
		MCWB	58.2	59.0	61.2	65.1	71.3	75.4	77.9	77.6	74.0	69.4	63.6	63.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	68.7	72.2	75.5	79.2	85.5	86.6	83.1	81.5	78.4	73.2	70.8
		MCDB	73.3	74.2	77.8	82.8	85.7	94.8	93.8	90.5	87.2	80.6	75.7	74.1
	2%	WB	66.3	66.1	68.9	73.0	76.1	82.9	83.7	81.4	79.5	76.2	70.2	69.0
		MCDB	70.5	72.8	74.6	79.0	83.0	91.2	91.3	87.0	83.4	80.8	72.3	72.1
	5%	WB	63.1	63.6	65.9	71.3	74.4	80.2	81.4	80.4	78.0	74.0	67.7	66.2
		MCDB	65.0	68.7	72.7	77.7	81.5	85.5	86.4	85.6	82.5	79.3	70.5	69.5
	10%	WB	59.1	60.7	63.4	68.8	73.0	77.9	79.9	79.1	76.2	72.2	64.3	63.0
		MCDB	63.2	65.7	68.9	74.2	79.8	84.2	85.6	84.0	82.5	76.8	67.8	65.1
Mean Daily Temperature Range	5% DB	MDBR	20.9	22.3	23.4	24.4	21.3	20.2	19.2	18.1	19.8	22.7	23.4	20.0
		MCDBR	25.0	26.2	27.4	27.3	25.4	22.3	22.4	20.7	23.0	24.3	25.7	22.9
		MCWBR	16.5	16.2	13.9	12.4	9.2	8.0	8.4	8.7	9.8	12.7	16.5	16.5
	5% WB	MCDBR	20.1	23.0	22.8	20.7	19.2	21.2	18.8	17.2	18.3	18.4	19.2	20.0
MCWBR		15.8	16.5	13.6	11.7	8.9	9.9	9.4	9.0	9.1	11.4	15.3	15.9	
Clear-Sky Solar Irradiance	taub		0.322	0.329	0.364	0.406	0.473	0.499	0.546	0.517	0.433	0.393	0.345	0.320
	taud		2.511	2.484	2.414	2.283	2.145	2.120	2.007	2.079	2.307	2.388	2.482	2.550
	Ebn at Noon		283	292	289	280	261	253	241	246	263	266	272	278
	Edh at Noon		27	30	35	42	49	50	56	51	39	33	27	24
All-Sky Solar Radiation	RadAvg		884	1086	1441	1826	1969	2030	1946	1752	1506	1269	976	783
	RadStd		49	111	119	124	151	141	121	89	127	148	76	65

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+329	+236

Nomenclature: See separate page